MIL-P-45621 (Ord)
7 May 1959
SUPERSEDING
FCS-857
23 April 1956

MILITARY SPECIFICATION

PANEL, RELAY, PLOTTING DATA SELECTION, 7621552

1. SCOPE

1.1 This specification covers one type of relay panel designated Panel, Relay, Plotting Data Selection, hereinafter referred to as "relay panel". The relay panel switches the inputs of the summing amplifiers for the horizontal range and altitude plotting boards. The relay panel is a component of the computers used with the Antiaircraft Fire Control Systems M33C and M33D.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

Federal

UU-T-111	Tape, Paper, Gummed
PPP-B-601	Boxes, Wood, Cleated-Plywood
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner
PPP-B-636	Boxes, Fiber
PPP-T-76	Tape, Pressure-Sensitive Adhesive, Paper, Water Resistant

Military

MIL-P-116	Preservation, Methods of

FSC 1285

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MIL-B-131	Barrier Material, Water Vaporproof, Flexible
MIL-D-3464	Desiccants, Activated, Bagged, Packaging Use and Static Dehumidification
MIL-C-7769	Cushioning Material, Bound
MIL-F-13926	Fire Control Materiel: General Specification Governing the Manufacture and Inspection of
MIL-P-14232	Parts, Equipment and Tools for Ordnance Materiel, Packaging of
MIL-F-45133	Fire Control Components and Sub-assemblies

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-202	Test Methods for Electronic and Electrical Component Parts
MS 20003	Indicator-Humidity, Card, Three Spot Impregnated Areas

DRAWINGS

Ordnance Corps

D7608180	Null Voltage Test Set
F7620834	Panel, Relay, Plotting Data Selection, Schematic Diagram
E7621552	Panel, Relay, Plotting Data Selection

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. - The following documents form a part of this specification. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

Consolidated Freight Classification Ratings, Rules and Regulations (Application for copies should be addressed to the Consolidated Freight Classification Committee, 202 Union Station, Chicago 6, Illinois.)

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National Motor Freight Classification No. A-2 and No. 13
(Application for copies should be addressed to the American Trucking Association, 1424 Sixteenth Street, N.W., Washington 6, D.C.)

3. REQUIREMENTS

3.1 Standard sample. - When specified in the contract a standard sample of the relay panel will be made available to the contractor to demonstrate the minimum acceptable standard of materials, design, workmanship, and performance (see 6.2 and 6.3).

3.2 Preproduction sample. - The contractor shall deliver one relay panel in accordance with 6.4 for testing and evaluation to prove that the relay panel is capable of meeting all requirements specified herein. The relay panel shall be a model representative of production.

3.3 Materials. - Materials shall be in accordance with the drawings, material specifications, and general specifications forming a part of this specification.

3.4 Design and construction. - The Panel, Relay, Plotting Data Selection shall be manufactured in accordance with Ordnance Corps Drawing E7621552, Panel, Relay, Plotting Data Selection, and drawings pertaining thereto. The requirements of this specification are detailed only to the extent considered necessary to insure the inherent quality and the performance capabilities of the product prior to final acceptance by the Government.

3.5 General specification. - The relay panel shall meet the following requirements of Specifications MIL-F-45133 and MIL-F-13926.

3.5.1 MIL-F-45133.

a. Tolerances
b. Functional or performance requirements
c. Incompatability of requirements

3.5.2 MIL-F-13926.

a. Order of precedence
b. General responsibilities of contractor
c. Drawings
d. Parts and material
e. Finish, mechanical
f. Finish, painting and protective
g. Lubrication
h. Electrical

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i. Identification of parts, assemblies, and complete items
j. Workmanship

3.6 Environmental.

3.6.1 Operating temperatures. - The relay panel shall meet all the performance requirements of this specification while exposed to, and thermally stabilized at, ambient temperatures of plus 150 degrees Fahrenheit (plus 65 degrees centigrade) and minus 40 degrees Fahrenheit (minus 40 degrees centigrade).

3.6.2 Storage temperatures. - The relay panel shall show no evidence of physical failure and shall meet all performance requirements of this specification at room temperature (plus 60 to 90 degrees Fahrenheit) after having been exposed to, and thermally stabilized at, ambient temperatures of plus 160 degrees Fahrenheit (plus 71 degrees centigrade) and minus 80 degrees Fahrenheit (minus 62 degrees centigrade).

3.6.3 Vibration. - The relay panel shall not be damaged or adversely affected and shall meet all the requirements of this specification after being subjected to a constant acceleration of 2.5 gravitational units during vibration of 3 minutes at each of the four most severe resonant frequencies. The four most severe resonant frequencies shall be determined by proper instrumentation during exposure to two vibratory cycles of 15-minute durations each over the frequency range of 5 to 500 cycles per second.

3.7 Electrical.

3.7.1 Continuity. - The relay panel shall demonstrate continuity in accordance with electrical schematic, Drawing F7620834.

3.7.2 Dielectric strength. - There shall be no puncture of insulation no sparkover, or decrease in dielectric strength when the insulated areas of the relay panel are subjected to an alternating-current (a-c) root mean square (rms) potential of 1000 volts plus twice the rated voltage at a frequency of 60 cycles per second for a period of not more than 30 seconds nor less than 5 seconds.

3.7.3 Insulation resistance. - Unless otherwise specified on applicable drawings, all insulation between ungrounded circuits and ground, and between isolated circuits, shall have a minimum resistance of 100 megohms with 500 volts direct-current (d-c) applied.

3.8 Performance. - When set up in accordance with pretest information, specified in section 4, the relay panel shall meet the following requirements. (All symbols referred to are those shown on Drawing F7620834 or on figure 1.)

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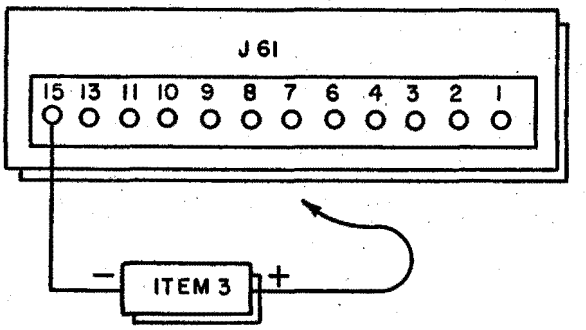
3.8.1 Relay operation.

a. Relay K61 shall energize when the positive terminal of a minus 28-volt power supply (figure 1A) is connected to terminal 3 of J61.
b. Relay K63 shall energize when the positive terminal of a minus 28-volt power supply is connected to terminal 2 of J61.
c. Relay K64 shall energize when the positive terminal of a minus 28-volt power supply is connected to terminal 4 of J61.
d. Relay K65 shall energize when the positive terminal of a minus 28-volt power supply is connected to terminal 6 of J61.
e. Relay K66 shall energize when the positive terminal of a minus 28-volt power supply is connected to terminal 8 of J61.
f. Relay K67 shall energize when the positive terminal of a minus 28-volt power supply is connected to terminal 10 of J61.
g. Relay K68 shall energize when the positive terminal of a minus 28-volt power supply is connected to terminal 7 of J61.
h. Relay K69 shall energize when the positive terminal of a minus 28-volt power supply is connected to terminal 9 of J61.
i. Relay K70 shall energize when the positive terminal of a minus 28-volt power supply is connected to terminal 11 of J61.
j. Relay K71 shall energize when the positive terminal of a minus 28-volt power supply is connected to terminal 1 of J61.
k. Relay K62 shall energize as relay K71 energizes when the positive terminal of a minus 28-volt power supply is connected to terminals 1 and 13 of J61.

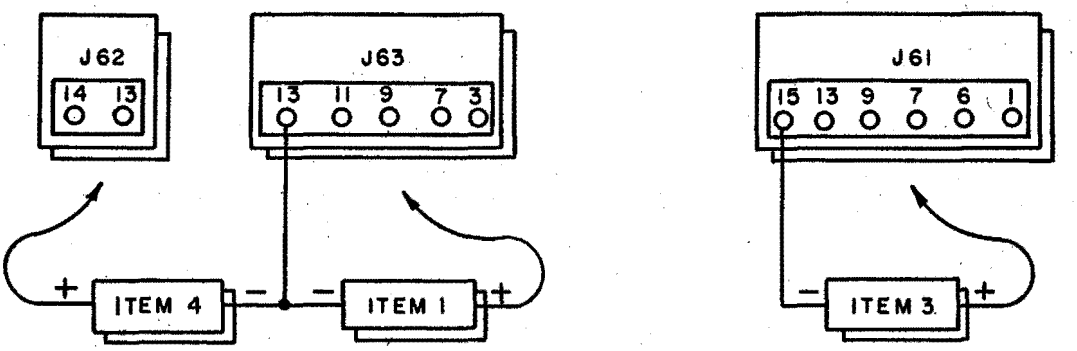
3.8.2 Voltage measurements.

a. The potential between terminals 7 and 13 of J63 shall be plus 5.6 plus or minus 0.6 volts when relay K68 energizes and the positive terminal of the minus 250-volt power supply is connected to terminal 14 of J62 (figure 1B).
b. The potential between terminals 11 and 13 of J63 shall be plus 5.6 plus or minus 0.6 volts when relays K62, K65, and K71 energize and the positive terminal of the minus 250-volt power supply is connected to terminal 14 of J62.

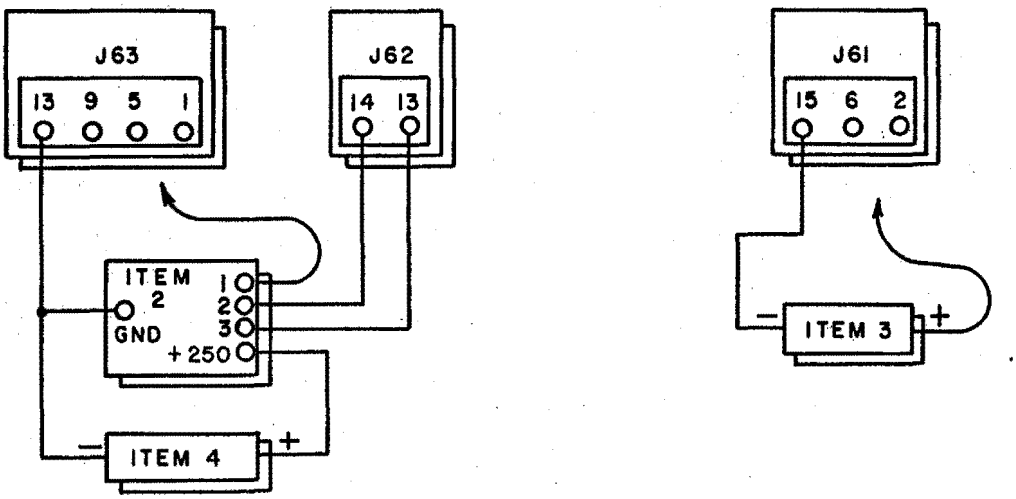
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A



B



C

Figure 1. Test circuits.

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c. The potential between terminals 9 and 13 of J63 shall be plus 0.1 plus or minus 0.05 volts when relays K62, K65, and K71 energize and the positive terminal of the minus 250-volt power supply is connected to terminal 14 of J62.
d. The potential between terminals 3 and 13 of J63 shall be plus 6.3 plus or minus 0.6 when relays K68 and K69 energize and the positive terminal of the minus 250-volt power supply is connected to terminal 13 of J62.

3.8.3 Resistance ratio measurements.

a. The ratio of the resistance between terminals 5 and 13 of J63 (figure 1C) to the resistance between terminal 14 of J62 and terminal 13 of J63 shall be 0.511 plus or minus 0.015 when relay K63 energizes.
b. The ratio of the resistance between terminals 9 and 13 of J63 to the resistance between terminal 14 of J62 and terminal 13 of J63 shall be 0.511 plus or minus 0.015 when relay K65 energizes.
c. The ratio of the resistance between terminals 1 and 13 of J63 to the resistance between terminal 13 of J62 and terminal 13 of J63 shall be 0.440 plus or minus 0.015 when relay K63 energizes.

4. QUALITY ASSURANCE PROVISIONS

4.1 General quality assurance provisions.

4.1.1 Contractor inspection. - Unless otherwise specified herein, the supplier is responsible for the performance of all inspection requirements prior to submitting the product for Government inspection and acceptance. Except as otherwise specified, the supplier may use his own facilities or any commerical laboratory acceptable to the Government. Complete inspection records of the examinations and tests shall be kept and shall be made available to the Government as specified in the contract or order.

4.1.2 Contractor quality assurance system. - The contractor shall provide and maintain an effective quality assurance system acceptable to the Government covering the supplies under this contract. A current written description of the system shall be submitted to the contracting officer prior to initiation of production. The written description will be considered acceptable when, as a minimum, it provides the quality assurance required by the specification and other documents referenced in this specification. The contractor will not be restricted to the inspection station or to the method of inspection listed, provided the substitute procedures will assure, as a minimum, the quality required in the contractual documents. In cases of dispute as to whether substitute procedures

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provide equal assurance, the comparable procedure of this specification shall be used. The contractor shall notify the Government of, and obtain approval for, any change to the written procedure that affects the degree of assurance required by this specification or other documents referenced herein.

4.1.3 Government verification. - All quality assurance operations performed by the contractor will be subject to Government verification at unscheduled intervals. Verification will consist of surveillance of the operations to determine whether the practices, methods, and procedures of the written inspection plan are being properly applied, and Government product inspection, when deemed necessary by the Government, to measure the quality of the product offered for acceptance. Deviation from the prescribed or agreed-upon procedures, or instances of poor practices which might have an effect upon the quality of the product, will immediately be called to the attention of the contractor. Failure of the contractor to promptly correct deficiencies discovered will be cause for suspension of acceptance until correction has been made or until conformance of product to prescribed criteria has been demonstrated. To avoid interference with operations, the contractor will designate a responsible official or officials to whom the Government inspector will report such instances.

4.2 Preproduction inspection.

4.2.1 Preproduction sample. - The preproduction samples specified in 3.2 shall be inspected by Frankford Arsenal for conformance to the requirement outlined herein. Preproduction samples shall be representative of the manufacturer's normal production.

4.2.2 Disposition of rejected preproduction samples. - Should the preproduction sample fail to pass the test for Government acceptance, production of those components causing rejection will cease until the defective characteristics have been corrected in the process used to produce the preproduction sample. When corrections in the process have been made, a new preproduction sample shall be completely inspected by the contractor and witnessed by the Government inspector.

4.3 Inspection provisions. - Quality assurance provisions specified in Specifications MIL-F-13926 and MIL-F-45133 shall apply where applicable and are not in conflict with the provisions specified herein.

4.3.1 Control of materials. - Materials shall be tested to determine conformance to contractually applicable specifications. Such tests may be conducted in the prime contractor's or subcontractor's facilities, or in other suitably equipped laboratories. The test results shall be available to the Government inspector for review. Certified test reports, identifiable with the material, may be accepted in lieu of repeated tests. The Government reserves the right, in any case, to require retests to assure that the materials are acceptable.

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4.3.2 Lot formation. - Unless otherwise specified by the contracting officer, or his representative, inspection lot sizes and lot formation shall be in accordance with Standard MIL-STD-105.

4.3.3 Examination. - The classification of defects in Tables I and II shall constitute the minimum inspection requirements to be performed prior to acceptance or rejection by lot. The Government reserves the right to inspect for any applicable requirement and to reject individual non-conforming items.

4.3.3.1 All rejected lots shall be screened for all defective characteristics. After corrections have been made and the Government inspector has been informed of the corrective action taken, the lot shall be resubmitted for acceptance inspection as outlined in Standard MIL-STD-105.

TABLE I - CLASSIFICATION OF DEFECTS

Normal Inspection, Level L7, and Sampling Plan Table IV-A of Standard MIL-STD-105
Reduced Inspection, Level L4, and Sampling Plan Table IV-A of Standard MIL-STD-105
(AQL* values are expressed in percent defective)

MAJOR	AQL 4.0	Req. Par.	QA Par.	Test Method
1.	Operating temperatures	3.6.1	4.4.1.1	SME**
2.	Storage temperatures	3.6.2	4.4.1.2	SME
3.	Vibration	3.6.3	4.4.1.3	SME

(*AQL - Acceptable Quality Level)
(**SME - Standard Measuring Equipment)

TABLE II - CLASSIFICATION OF DEFECTS

Normal Inspection, Level II, and Sampling Plan Table IV-A of Standard MIL-STD-105
Reduced Inspection, Level L7, and Sampling Plan Table IV-A of Standard MIL-STD-105
(AQL values are expressed in percent defective)

MAJOR	AQL 1.5	Req. Par.	QA Par.	Test Method
1.	Completeness of assembly		As per drawings	Visual
2.	Continuity	3.7.1	4.4.2.1	SME
3.	Dielectric strength	3.7.2	4.4.2.2	SME
4.	Insulation resistance	3.7.3	4.4.2.3	SME
5.	Relay operation	3.8.1	4.4.3.2	SME

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TABLE II - CLASSIFICATION OF DEFECTS (Cont'd)

MAJOR	AQL 1.5	Req. Par.	QA Par.	Test Method
6.	Voltage measurements	3.8.2	4.4.3.3	SME
7.	Resistance ratio measurements	3.8.3	4.4.3.4	SME

4.3.4 Inspection equipment. - The test methods and procedures outlined in 4.4 shall be used as standards for determining the adequacy of the contractor's inspection equipment and methods used for inspecting the characteristics in Tables I and II. Unless otherwise specified in the contract, the contractor shall provide and maintain gages and other measuring and testing devices considered necessary for inspection of supplies for conformance to contractual requirements. The Government reserves the right to certify all such equipment and to use this equipment for Government verification inspection.

4.4 Test methods and procedures.

4.4.1 Environmental.

4.4.1.1 Operating temperatures. - Unless otherwise specified, the test equipment to be used when performing the temperature tests shall be such that the equipment under test shall not exceed fifty (50) percent of the internal volume of the test chamber, and no portion shall be closer than six (6) inches from any surface of the chamber.

4.4.1.1.1 The relay panel to be tested shall be placed in the test chamber in its normal operating position. The test shall be conducted in accordance with Procedure I of Specification MIL-F-13926 to the conditions specified in 3.6.1. Upon completion of the specified exposures, the unit under test shall be inspected to determine conformance with the requirements specified in 3.6.1.

4.4.1.2 Storage temperature. - The test shall be conducted in accordance with Procedure II of Specification MIL-F-13926 to the conditions specified in 3.6.2. Upon completion of the specified exposure, the unit under test shall be inspected to determine conformance with the requirements specified in 3.6.2.

4.4.1.3 Vibration. - The relay panel shall be mounted, using its own mounting provisions, on a vibrating machine capable of producing a motion at a constant frequency within the frequency range specified in 3.6.3. The testing machine shall be equipped with a medium to insure that the mounting is free from resonances over the test frequency range. The medium shall hold the unit under test in its normal operating position. The measuring instruments of the testing equipment shall be capable of measuring the frequency within plus or minus 2 percent.

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4.4.1.3.1 The unit under test shall be vibrated successively along each of its three (3) mutually perpendicular axes that are respectively parallel to the edges of the unit with an acceleration, duration, and frequency conforming to the conditions specified in 3.6.3.

4.4.1.3.2 Resonant frequencies of the unit under test shall be determined by varying the frequency of applied vibration slowly through the specified frequency range at a vibratory acceleration not exceeding the specified value in 3.6.3. In no instance shall the unit under test be vibrated at any resonant mode for periods less than the period specified in 3.6.3. Upon completion of the test, the unit shall be inspected to determine that the requirements specified in 3.6.3 are met.

4.4.2 Electrical.

4.4.2.1 Continuity. - A continuity check of the relay panel circuits shall be made, using standard measuring devices to determine conformance with the requirements specified in 3.7.1.

4.4.2.2 Dielectric strength. - The unit shall be tested in accordance with Method 301 of Standard MIL-STD-202, and shall meet the requirements specified in 3.7.2.

4.4.2.3 Insulation resistance. - Immediately after satisfactorily withstanding the test for dielectric strength, the unit shall be given the insulation resistance test in accordance with Method 203, Test Condition B, of Standard MIL-STD-202, and shall meet the requirements specified in 3.7.3.

4.4.3 Performance.

4.4.3.1 Pretest information.

4.4.3.1.1 Test equipment listed in Table III shall be connected as shown on figure 1, or as specified in section 4.4.3.

TABLE III - TEST EQUIPMENT

Item Number	Description
1	Vacuum-tube voltmeter Voltage range: 50 millivolts to 1200 volts d-c, 100 millivolts to 1200 volts a-c. Accuracy: plus or minus 5 percent of the full scale value. Frequency range for ac: 30 cycles to 3 megacycles per second.

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TABLE III - TEST EQUIPMENT (Cont'd)

Item Number	Description
	Input impedance: 1 megohm for d-c measurements, 2 megohms for a-c measurements. Power supply: 115/230 volts plus or minus 10 percent, 50 to 420 cycles per second.
2	Null voltage test set: per Ordnance Corps Drawing D7608180.
3	Power supply: minus 28-volt d-c, 1.0-ampere.
4	Power supply: plus 250-volt d-c plus or minus 10 volts, 0.01-ampere.

4.4.3.1.2 Unless otherwise specified, all adjustments are made to the test equipment (figure 1) and not to the unit under test.

4.4.3.2 Relay operation. - Item 3 shall be connected as shown on figure 1A for each of the following requirements:

a. The positive terminal of item 3 shall be connected to terminal 3 of J61. Relay K61 shall perform as specified in 3.8.1a.
b. The positive terminal of item 3 shall be connected to terminal 2 of J61. Relay K63 shall perform as specified in 3.8.1b.
c. The positive terminal of item 3 shall be connected to terminal 4 of J61. Relay K64 shall perform as specified in 3.8.1c.
d. The positive terminal of item 3 shall be connected to terminal 6 of J61. Relay K65 shall perform as specified in 3.8.1d.
e. The positive terminal of item 3 shall be connected to terminal 8 of J61. Relay K66 shall perform as specified in 3.8.1e.
f. The positive terminal of item 3 shall be connected to terminal 10 of J61. Relay K67 shall perform as specified in 3.8.1f.
g. The positive terminal of item 3 shall be connected to terminal 7 of J61. Relay K68 shall perform as specified in 3.8.1g.
h. The positive terminal of item 3 shall be connected to terminal 9 of J61. Relay K69 shall perform as specified in 3.8.1h.
i. The positive terminal of item 3 shall be connected to terminal 11 of J61. Relay K70 shall perform as specified in

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5.1.1.3.1 Immobilizing in fiberboard box. - The unit shall be immobilized within a fiberboard box by slotted support blocks at either end. (See figure 2.) The fiberboard box shall be in accordance with Specification PPP-B-636, class 1, style RSC, 125 pound test. The size of the wood support blocks and the fiberboard box shall be based upon the size of the assembly. The height of the container shall be sufficient to permit a minimum clearance of 1/2 inch from the top and bottom of the electronic unit when it is positioned in the container. The length of the fiberboard box shall be such as to provide a snug fit against the support blocks when the unit is in position.

5.1.1.3.2 Desiccant. - The required units of desiccant, conforming to Specification MIL-D-3464, computed according to Formula I of Specification MIL-P-116, shall be secured to the assembly.

5.1.1.3.3 Closure of fiberboard container. - The fiberboard box shall be closed with gummed tape conforming to Specification UU-T-111.

5.1.1.3.4 Humidity indicator. - A humidity indicator, in accordance with Standard MS 20003, shall be secured to the top surface of the fiberboard box.

5.1.1.3.5 Barrier. - The corners of the box shall be blunted and the box shall be sealed within a barrier fabricated of material conforming to Specification MIL-B-131, class 1. The time lapse between securing the desiccant and the closure of the barrier shall be held to an absolute minimum.

5.1.1.3.6 Cushioning. - The unit shall be cushioned within an outer fiberboard box. Cushioning material shall be a minimum thickness of one inch on all surfaces and shall conform to type II of Specification MIL-C-7769.

5.1.1.3.7 Outer fiberboard box. - The outer fiberboard box shall conform to Specification PPP-B-636, class 2, W5c, style RSC.

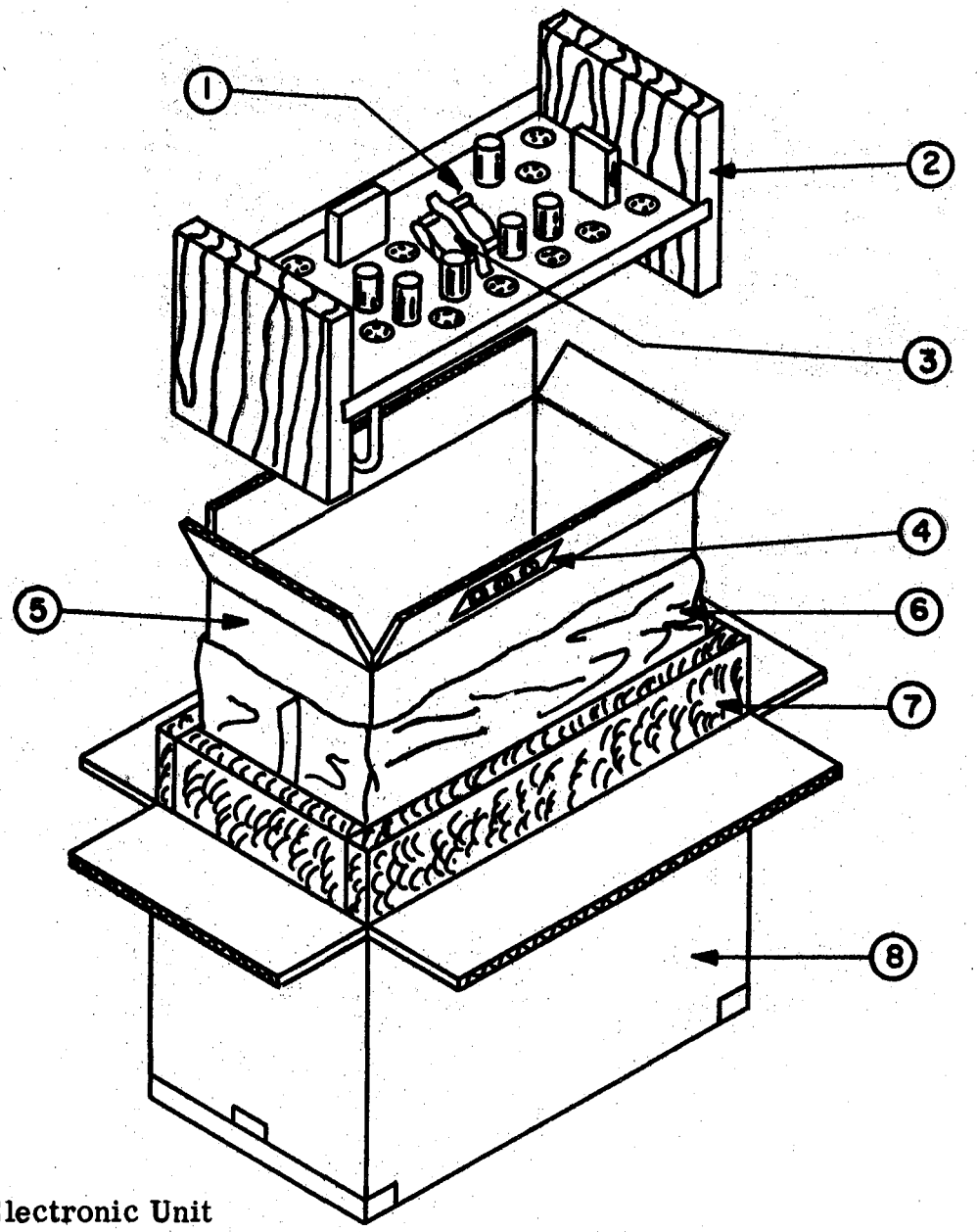
5.1.1.3.8 Closure of container. - The fiberboard box shall be sealed with pressure sensitive tape conforming to Specification PPP-T-76 (minimum width 2 inches) in accordance with requirements of Specification PPP-B-636.

5.1.2 Level C. - The unit shall be preserved and packaged in accordance with commercial practice. The packaging shall protect the unit from deterioration during shipment and during the storage period prior to use.

5.2 Packing.

5.2.1 Level A. - A quantity of packaged units (not exceeding 12) shall be immobilized in a shipping container conforming to either of the following specifications:

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1. Electronic Unit
2. Support Blocks - group I or II lumber of Spec. PPP-B-621 - 2-req'd.
3. Desiccant conforming to Spec. MIL-D-3464.
4. Humidity indicator - MS 20003.
5. Inner fiberboard box - type I, classI, style RSC of Spec. PPP-B-636.
6. Water vaporproof barrier - class I of Spec. MIL-B-131.
7. Bound cushioning material - MIL-C-7769.
8. Outer fiberboard box - W5c, style RSC of Spec. PPP-B-636.

Figure 2. Packaging.

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PPP-B-601, overseas type
PPP-B-621, class 2 - overseas

5.2.2 Level B. - A quantity of packaged units (not exceeding 12) shall be immobilized in a shipping container conforming to either of the following specifications:

PPP-B-601, domestic type
PPP-B-621, class 1 - domestic

5.2.3 Level C. - Packaged units shall be protected from damage by packing in conformance with Consolidated Freight Classification Rules and Container Specifications for Rail Shipment or with National Motor Freight Classification Rules and Container Specifications for Truck Shipment.

5.3 Marking. - Unit packages and shipping containers shall be marked in accordance with the requirements of Standard MIL-STD-129.

5.4 Pilot pack. - Prior to quantity packaging, a pilot pack shall be inspected to determine conformance with the applicable requirements of this specification. The pilot pack shall be a completely packed shipping container. Inspection may be performed either in process or after completion of the pilot pack.

5.5 Workmanship. - Adequate controls shall be used to check for contamination of cleaning solvents. Tape shall be applied uniformly to interior boxes and shall completely cover and adhere to all joints and seams. Shipping containers shall be free of protruding staples, screws, nails, etc. Strapping shall be applied just prior to shipment and shall not cover any container marking. Irrelevant marking shall be neatly obliterated.

6. NOTES

6.1 Intended use. - The plotting data selection relay panel is used to switch the input circuits of the servo summing amplifiers of the horizontal range plotting board and the altitude plotting board. The switches of the unit control the signal voltages, reference mark voltages, offset voltages, and standby voltages, and provide for interchanging the proper data to accomplish the function of pen interchange.

6.2 Ordering data. - Procurement documents should specify the following:

a. Title, number, and date of this specification.
b. Selection of applicable levels of packaging and packing.
c. Applicable stock number.
d. Availability of standard sample as specified in 3.1.

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e. Provisions for preproduction sample(s) including date of delivery. (See 3.2 and 6.4.)

6.3 Standard sample. - A standard sample is a sample essential to illustrate requirements of the specification and is not to be used as a standard of calibration.

6.4 Preproduction sample delivery. - The preproduction sample shall be delivered by the contractor to Frankford Arsenal, Bridge and Tacony Sts., Phila. 37, Pa. attn: ORDBA-QA-5300 within the time designated by the contract. (See 6.2.)

6.5 Design information. - Design information regarding specific equipment not covered or referenced by this specification may be obtained from the procuring activity upon the specific request of the contractor.

6.6 Definitions.

6.6.1 Standard measuring equipment (SME). - Standard measuring equipment (SME) includes the common, hand-type measuring devices which are usually stocked by commercial supply houses for ready supply (shelf items), and which are normally used by an inspector to perform dimensional inspection of items under procurement. This category also includes commercial testing equipment.

NOTICE: When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any invention that may in any way be related thereto.

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3.8.1i.

j. The positive terminal of item 3 shall be connected to terminal 1 of J61. Relay K71 shall perform as specified in 3.8.1j.

k. The positive terminal of item 3 shall be connected to terminals 1 and 13 of J61. Relays K62 and K71 shall perform as specified in 3.8.1k.

4.4.3.3 Voltage measurements. - Items 1, 3, and 4 shall be connected as shown on figure 1B for each of the following requirements.

a. The positive terminal of item 4 shall be connected to terminal 14 of J62. The positive terminal of item 1 shall be connected to terminal 7 of J63. The positive terminal of item 3 shall be connected to terminal 7 of J61, energizing relay K68. The indication on the voltmeter, item 1, shall be as specified in 3.8.2a.

b. The positive terminals of items 1 and 3 shall be removed from J63 and J61. The positive terminal of item 1 shall be connected to terminal 11 of J63. The positive terminal of item 3 shall be connected to terminals 1, 6, and 13 of J61, energizing relays K62, K65, and K71. The indication on the voltmeter, item 1, shall be as specified in 3.8.2b.

c. The positive terminal of item 1 shall be removed from terminal 11 of J63 and connected to terminal 9 of J63. The indication on the voltmeter, item 1, shall be as specified in 3.8.2c.

d. The positive terminals of items 1, 3, and 4 shall be removed from J63, J61, and J62. The positive terminal of item 4 shall be connected to terminal 13 of J62. The positive terminal of item 1 shall be connected to terminal 3 of J63. The positive terminal of item 3 shall be connected to terminal 7 and 9 of J61, energizing relays K68 and K69. The indication on the voltmeter, item 1, shall be as specified in 3.8.2d.

e. The positive terminals of items 1, 3, and 4 shall be removed from J63, J61, and J62.

4.4.3.4 Resistance ratio measurements. - Items 2, 3, and 4 shall be connected as shown on figure 1C for each of the following requirements.

a. The positive terminal of item 3 shall be connected to terminal 2 of J61, energizing relay K63. The TEST 1 terminal of item 2 shall be connected to terminal 5 of J63.

b. The resistance between terminal 5 of J63 and ground shall be determined and noted. The resistance between

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terminal 14 of J62 and ground shall be determined and noted. The ratio of the two resistance values shall be as specified in 3.8.3a.

c. The positive terminal of item 3 shall be removed from terminal 2 of J61 and connected to terminal 6 of J61, energizing relay K65. The TEST 1 terminal of item 2 shall be removed from terminal 5 of J63 and connected to terminal 9 of J63.

d. The resistance between terminal 9 of J63 and ground shall be determined and noted. The resistance between terminal 14 of J62 and ground shall be determined and noted. The ratio of the two resistance values shall be as specified in 3.8.3b.

e. The positive terminal of item 3 shall be removed from terminal 6 of J61 and connected to terminal 2 of J61, energizing relay K63. The TEST 1 terminal of item 2 shall be removed from terminal 9 of J63 and connected to terminal 1 of J63.

f. The resistance between terminal 1 of J63 and ground shall be determined and noted. The resistance between terminal 13 of J62 and ground shall be determined and noted. The ratio of the two resistance values shall be as specified in 3.8.3c.

4.5 Inspection of components and subassemblies. - Components and subassemblies shall be inspected in accordance with Specification MIL-F-45133.

4.6 Packaging. - Quality assurance for packaging shall be as specified in Specification MIL-P-14232.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.

5.1.1 Level A.

5.1.1.1 Cleaning. - The unit shall be cleaned in accordance with process C-1 of Specification MIL-P-116.

5.1.1.2 Drying. - Drying shall be accomplished by procedure D-1 of Specification MIL-P-116.

5.1.1.3 Unit packaging. - The item shall be packaged method IIb, in accordance with Specification MIL-P-116, and as specified herein.

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